

Exponential Family Regularity Conditions: Derivation of $E[a(Y)]$ and $\text{Var}[a(Y)]$

Dominion

Setup

We work with a one-parameter exponential family density written in the form

$$f(y; \theta) = \exp [a(y)b(\theta) + c(\theta) + d(y)].$$

Here $a(y)$ and $d(y)$ depend only on the data, while $b(\theta)$ and $c(\theta)$ depend only on the parameter. The two “regularity conditions” below come from differentiating

$$\int f(y; \theta) dy = 1$$

with respect to θ , once to get $E[a(Y)]$, and twice to get $\text{Var}[a(Y)]$. Differentiating under the integral sign is valid under standard regularity conditions on f .

1 First Derivative: $E[a(Y)]$

Since $\int f(y, \theta) dy = 1$ for every θ , differentiating both sides with respect to θ gives

$$\frac{d}{d\theta} \int f(y, \theta) dy = 0.$$

Assuming we may swap the order of differentiation and integration,

$$\int \frac{df(y; \theta)}{d\theta} dy = 0. \tag{1}$$

Now differentiate $f(y; \theta) = \exp[a(y)b(\theta) + c(\theta) + d(y)]$ with respect to θ . Since $a(y)$ and $d(y)$ do not depend on θ , only $b(\theta)$ and $c(\theta)$ contribute inside the exponent, via the chain rule:

$$\frac{d}{d\theta} \left[\exp\{a(y)b(\theta) + c(\theta) + d(y)\} \right] = \exp [a(y)b(\theta) + c(\theta) + d(y)] \times [a(y)b'(\theta) + c'(\theta)].$$

The exponential factor is just $f(y; \theta)$ again, so

$$\frac{df(y; \theta)}{d\theta} = f(y; \theta) [a(y)b'(\theta) + c'(\theta)]. \tag{2}$$

Substituting (2) into (1):

$$\int a(y)b'(\theta) f(y; \theta) dy + \int c'(\theta) f(y; \theta) dy = 0.$$

Since $b'(\theta)$ and $c'(\theta)$ do not depend on y , they factor out of their respective integrals:

$$b'(\theta) \int a(y) f(y; \theta) dy + c'(\theta) \int f(y; \theta) dy = 0.$$

The first integral is $E[a(Y)]$ by definition. The second integral is $\int f(y; \theta) dy = 1$, since f is a proper density. So

$$b'(\theta) E[a(Y)] + c'(\theta) \cdot 1 = 0.$$

$$\boxed{E[a(Y)] = -\frac{c'(\theta)}{b'(\theta)}} \quad (3)$$

This is the first Bartlett-type identity for the exponential family: it lets you read off $E[a(Y)]$ directly from the natural parameter functions $b(\theta)$ and $c(\theta)$, without ever computing the integral $\int y f(y; \theta) dy$ directly.

Worked example: Binomial case

For $Y \sim \text{Binomial}(n, \pi)$, writing the pmf in exponential family form with natural parameter $\theta = \ln\left(\frac{\pi}{1-\pi}\right)$ gives

$$b(\theta) = \ln\left(\frac{\pi}{1-\pi}\right), \quad c(\theta) = n \ln(1-\pi).$$

Applying (3) with these choices (differentiating with respect to π and using the chain rule to convert back to θ) gives

$$E[a(Y)] = -\frac{c'(\theta)}{b'(\theta)} = \frac{n}{1-\pi} \bigg/ \frac{1}{\pi(1-\pi)} = n\pi.$$

This matches the familiar result $E[Y] = n\pi$ for the binomial distribution, confirming $a(y) = y$ here, exactly as expected.

2 Second Derivative: $\text{Var}[a(Y)]$

Now differentiate $\int f(y, \theta) dy = 1$ *twice* with respect to θ :

$$\frac{d^2}{d\theta^2} \int f(y, \theta) dy = 0 \quad \Longleftrightarrow \quad \frac{d}{d\theta} \int \frac{d}{d\theta} f(y, \theta) dy = 0. \quad (4)$$

We already found in (2) that

$$\frac{d}{d\theta} f(y; \theta) = f(y; \theta) [a(y)b'(\theta) + c'(\theta)],$$

so (4) becomes

$$\int \frac{d}{d\theta} \left\{ f(y; \theta) [a(y)b'(\theta) + c'(\theta)] \right\} dy = 0. \quad (5)$$

The integrand is a product of two θ -dependent factors, $f(y; \theta)$ and $g(\theta) := a(y)b'(\theta) + c'(\theta)$. By the product rule,

$$\frac{d}{d\theta} [f(y; \theta) g(\theta)] = \underbrace{f'(y; \theta)}_{= f(y; \theta) g'(\theta)} g(\theta) + f(y; \theta) g'(\theta),$$

where we re-used (2) for $f'(y; \theta)$, and $g'(\theta) = a(y)b''(\theta) + c''(\theta)$. Hence

$$\frac{d}{d\theta} [f(y; \theta) g(\theta)] = f(y; \theta) [a(y)b'(\theta) + c'(\theta)]^2 + f(y; \theta) [a(y)b''(\theta) + c''(\theta)].$$

Substituting into (5):

$$\int f(y; \theta) [a(y)b'(\theta) + c'(\theta)]^2 dy + \int f(y; \theta) [a(y)b''(\theta) + c''(\theta)] dy = 0. \quad (6)$$

From (3), $c'(\theta) = -b'(\theta)E[a(Y)]$, so

$$a(y)b'(\theta) + c'(\theta) = b'(\theta)[a(y) - E[a(Y)]].$$

The first integral in (6) therefore becomes

$$\int f(y; \theta) [b'(\theta)]^2 [a(y) - E[a(Y)]]^2 dy = [b'(\theta)]^2 \int f(y; \theta) [a(y) - E[a(Y)]]^2 dy = [b'(\theta)]^2 \text{Var}[a(Y)],$$

since $\int f(y; \theta) [a(y) - E[a(Y)]]^2 dy$ is exactly the definition of $\text{Var}[a(Y)]$.

For the second integral in (6),

$$\int f(y; \theta) [a(y)b''(\theta) + c''(\theta)] dy = b''(\theta) \underbrace{\int a(y)f(y; \theta) dy}_{=E[a(Y)]} + c''(\theta) \underbrace{\int f(y; \theta) dy}_{=1} = b''(\theta)E[a(Y)] + c''(\theta).$$

Putting both pieces back into (6):

$$[b'(\theta)]^2 \text{Var}[a(Y)] + b''(\theta)E[a(Y)] + c''(\theta) = 0.$$

Substituting $E[a(Y)] = -c'(\theta)/b'(\theta)$ from (3):

$$[b'(\theta)]^2 \text{Var}[a(Y)] = -b''(\theta) \left(-\frac{c'(\theta)}{b'(\theta)} \right) - c''(\theta) = \frac{b''(\theta)c'(\theta)}{b'(\theta)} - c''(\theta).$$

Multiplying both sides by $b'(\theta)$ and rearranging:

$$b'(\theta) \left[[b'(\theta)]^2 \text{Var}[a(Y)] \right] = - \left[b''(\theta)c'(\theta) - c''(\theta)b'(\theta) \right] = c''(\theta)b'(\theta) - b''(\theta)c'(\theta).$$

Dividing through by $[b'(\theta)]^3$ gives the closed-form expression:

$$\boxed{\text{Var}[a(Y)] = \frac{c''(\theta)b'(\theta) - b''(\theta)c'(\theta)}{[b'(\theta)]^3}} \quad (7)$$

This is the exponential-family analogue of the Fisher Information relation: once you know $b(\theta)$ and $c(\theta)$, both moments $E[a(Y)]$ and $\text{Var}[a(Y)]$ follow directly from (3) and (7), without ever integrating the density explicitly.